

GraphRAG Architecture Document (GRAD)

IntelliPlant AI

AI-Powered Industrial Knowledge Intelligence Platform

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1. Introduction

GraphRAG (Graph Retrieval-Augmented Generation) combines:

- Vector Search
- Knowledge Graph Traversal
- Keyword Search
- Multi-Agent Reasoning
- LLM-based Context Synthesis

Unlike traditional RAG, GraphRAG leverages entity relationships and graph traversal to improve answer quality, explainability, and reasoning.

2. Why GraphRAG?

Traditional RAG limitations:

- ✗ Retrieves isolated chunks
 - ✗ No entity relationships
 - ✗ Weak multi-hop reasoning
 - ✗ Context fragmentation
 - ✗ Poor explainability
-

GraphRAG advantages:

- ✓ Multi-hop reasoning
 - ✓ Entity awareness
 - ✓ Context enrichment
 - ✓ Explainability
 - ✓ Better citations
 - ✓ Lower hallucinations
-

3. GraphRAG Architecture Overview

```text id="o3jdxu" User Query | ▼ Query Understanding | ▼ Hybrid Retrieval Layer  
----- Vector Search Graph  
Search BM25 (Qdrant) (Neo4j) Search  
----- | ▼ Context Fusion  
Engine | ▼ Graph Expansion Layer | ▼ Gemini | ▼ Citation Generator | ▼ Final Response

---

# 4. High-Level Architecture



Operator

Failure

Inspection

WorkOrder

Regulation

Plant

Sensor

---

## Relationships

CONNECTED\_TO

FAILED\_IN

CAUSES

DESCRIBED\_BY

REQUIRES

REFERENCES

SIMILAR\_TO

INSPECTED\_BY

---

Example

```text id="mz6e2j" Pump P101

↓

FAILED_IN

↓

Bearing Failure

↓

CAUSES

↓

Overheating

```
---  
  
# 6. Vector Retrieval Layer  
  
Technology  
  
Qdrant  
  
Purpose  
  
Semantic similarity search.  
  
---  
  
Embedding Model  
  
BGE-Large  
  
Dimensions  
  
1024  
  
---  
  
Stored Metadata  
  
```json id="rfrt5h"  
{
 "document_id": "",
 "chunk_text": "",
 "equipment": "Pump P101",
 "section": "Maintenance"
}
```

## 7. BM25 Retrieval Layer

Purpose

Keyword matching.

---

Useful For

Equipment IDs

Model numbers

Part numbers

Standards

Serial numbers

---

Example

```
```text id="pxylh4" P101
```

ISO 45001

OISD-STD-144

```
---
```

```
# 8. Hybrid Retrieval Layer
```

```
Combines:
```

```
### Vector Search
```

```
Semantic relevance.
```

```
Weight = 40%
```

```
---
```

```
### Graph Search
```

Relationship evidence.

Weight = 40%

BM25 Search

Keyword relevance.

Weight = 20%

Retrieval Formula

```text id="hj9f9z"

Final Score =

0.4 × Vector Score +

0.4 × Graph Score +

0.2 × BM25 Score

---

## 9. Query Processing Pipeline

```text id="8xj6ra" User Query

↓

Query Rewriter

↓

Intent Detection

↓

Entity Detection

↓

Hybrid Search

↓

Graph Expansion



Context Fusion



LLM



Response

```
---  
  
# 10. Entity Recognition  
  
Extract entities:  
  
Equipment  
  
Failure  
  
Operators  
  
Dates  
  
Regulations  
  
Incidents  
  
Examples  
  
```text id="x8o3lu"  
Pump P101

Bearing Failure

Operator A

ISO 9001
```



# 11. Graph Traversal

Example Query

"Why did Pump P101 fail?"

Traversal

```text id="bym4uc" Pump P101

↓

FAILED_IN

↓

Bearing Failure

↓

CAUSES

↓

Overheating

↓

SIMILAR_TO

↓

Incident #124

Multi-hop traversal enriches context.

12. Graph Expansion

Purpose

Discover neighboring knowledge.

Example

Pump P101

↓

Motor M201

↓

Inspection Report

↓

Operator B

↓

Maintenance Manual

Expanded context improves answer quality.

13. Context Fusion Engine

Inputs

Vector Context

Graph Context

Keyword Context

Memory Context

Process

Ranking

Deduplication

Compression

Relevance filtering

Output

Unified prompt context

14. GraphRAG Workflow

```text id="lvjz5u"

Query

↓

Intent Detection

↓

Entity Extraction

↓

Vector Search

↓

Graph Traversal

↓

BM25 Search

↓

Context Fusion

↓

LLM Reasoning

↓

Citation Generation

↓

Answer

---

## 15. Multi-Hop Reasoning

Traditional RAG

```text id="8j8j2v" Question

↓

Chunk

↓

Answer

GraphRAG

```text id="e0zok5"

Question

↓

Equipment

↓

Incident

↓

Failure

↓

Manual

↓

Recommendation

↓

Answer

---

## 16. Citation Engine

Sources include:

Manuals

Incident Reports

SOPs

Inspection Reports

Standards

---

Example

```
```json id="6vk0x9" { "answer":"","sources":["manual.pdf","incident_2024.pdf"] }
```

```
---
```

```
# 17. Confidence Scoring
```

```
Confidence Formula
```

```
```text id="g0l1xg"
```

```
Confidence =
```

```
0.3 Retrieval Score +
```

```
0.3 Graph Evidence +
```

```
0.2 Citation Quality +
```

```
0.2 LLM Consistency
```

---

Range

0-1

---

Example

0.92

---

## 18. Hallucination Prevention

Grounded generation.

Multiple retrieval methods.

Source citations.

Graph evidence.

Confidence threshold.

---

Below 0.7 confidence:

```text id="s0fjlm" Insufficient evidence available.

19. Memory Integration

Short-Term Memory

Redis

Stores:

Recent conversation.

Long-Term Memory

Neo4j

Stores:

Historical knowledge.

Semantic Memory

Qdrant

Stores:

Embeddings.

20. Example Workflow

Question

```
```text id="l3s1o3"
Why Pump P101 failed?
```

Process

```
```text id="ct2y9n" Detect Equipment
```

↓

Vector Search

↓

Find Similar Incidents

↓

Graph Traversal

↓

Retrieve Manual

↓

Find Root Cause

↓

Generate Recommendation

↓

Answer

```
---

# 21. Performance Metrics

| Metric | Target |
|-----|-----|
| Retrieval Precision | >90% |
| Citation Accuracy | >95% |
```

```
| Hallucination Rate | <5% |  
| Query Latency | <5 sec |  
| Graph Coverage | >85% |  
| Multi-hop Accuracy | >90% |
```

22. Failure Handling

Qdrant Down

↓

Neo4j + BM25

Neo4j Down

↓

Vector Search

LLM Failure

↓

Fallback Model

Graph Expansion Failure

↓

Direct Retrieval

23. Future Enhancements

Hierarchical GraphRAG

Graph Neural Networks

Knowledge Graph Embeddings

Self-Learning Graphs

Temporal GraphRAG

Digital Twin Integration

Predictive Maintenance GraphRAG

Agentic GraphRAG

Multimodal GraphRAG

24. Complete GraphRAG Pipeline

```text id="vrvq9v"

Document Upload

↓

OCR

↓

Chunking

↓

Embeddings

↓

Qdrant

↓

Entity Extraction

↓

Knowledge Graph

↓

Neo4j

↓

Hybrid Retrieval

↓

Context Fusion

↓

Gemini/OpenAI

↓

Citation Engine

↓

Final Answer

## Conclusion

GraphRAG is the core intelligence engine of IntelliPlant AI. By combining vector search, knowledge graphs, multi-hop reasoning, and explainable AI, the system transforms fragmented industrial documents into a contextual and trustworthy Industrial Operations Brain capable of delivering accurate, citation-backed, and low-hallucination answers.